

Vacuum arc melted and heat treated AlCoCrFeNiTi_x based high-entropy alloys: Thermodynamic and microstructural investigations



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ABSTRACT

The AlCoCrFeNi alloy system is one of the commonly studied high-entropy alloys (HEA). The additional elements like Ti affect the properties of HEAs. In order to predict these effects on the microstructure and the properties of HEAs, a suitable phase diagram is needed. The phase diagrams can be plotted by using thermodynamic software and databases. But experimental studies should be done to control whether this phase diagram is suitable or not. The aim of this study is to propose a phase diagram suitable for the AlCoCrFeNiTi_x system by using ThermoCalc software and to compare it with the experimental results. In this work, HEAs were produced by a vacuum arc melting (VAM) method followed by a heat treatment (H-T) process. The stoichiometric amount of Ti was varied ($X = 0.0, 0.2, 0.4, 0.6, 0.8, \text{ and } 1.0$). Its effects on the microstructure evaluations and phase transformations in HEAs were investigated. The H-T processes were realized at different temperatures and cooling conditions. The thermodynamic investigations were performed to predict the phases present within as-cast (AC) and heat-treated (HT) HEAs by using Thermo-Calc 2021a Software and its TCHEA4 database. The Calculation of Phase Diagram (CALPHAD) studies were also utilized to plot the phase diagrams for the AlCoCrFeNiTi_x system under both equilibrium and non-equilibrium cooling conditions. AC and HT HEAs were characterized by various techniques: scanning electron microscope (SEM) with an energy dispersive spectrometer (EDS), X-ray diffractometer (XRD), differential thermal analysis (DTA), and differential scanning calorimetry (DSC). The phases and the microstructures of AC HEAs were detected as similar to the theoretically obtained alloys under non-equilibrium cooling conditions. While the phases and the microstructures of HT HEAs were detected as similar to the theoretically obtained alloys under equilibrium cooling conditions. As a result, the target alloy composition and production process can be designed depending on the desired properties using the proposed phase diagrams. For example, if the Sigma phase is not desired in the structure, HEAs with Ti1.1 and above contents should be studied.

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1. Introduction

The definition of high-entropy alloys (HEAs) is based on the content of multiple principal elements that enables the stability of solid solution phases to be increased due to their higher mixing entropy values [1]. The production methods of HEAs are mainly depend on the processes where high speed solidification is realized such as, melt spinning, chill casting, the Bridgman-Stockbarger method, vacuum arc melting, etc [2]. In one of the frontier studies on

HEAs, Cantor et al. investigated the microstructural properties of HEAs, which contained 16 and 20 principal elements in equiatomic amounts, respectively, and was produced via induction melting and melt spinning. They obtained HEAs containing one main phase (FCC) with a small amount of interdendritic phases [3,4]. The mechanical properties such as, yield and tensile strength, plasticity, ductility, hardness of the HEAs are directly related to their crystal structures. Their compositions, cooling regimes and heat treatment processes can modify HEAs microstructures. In addition, the casting, cooling and heat treatment steps of HEAs can affect their electrical, magnetic, thermal and electrochemical properties [4]. These newly developed alloy series are used in many fields. Since their creation in the last decade, HEAs have accumulated considerable interest in the area of materials research owing to their high strength and hardness,

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superior softening resistance, wear and corrosion resistance, and unique ability to form phases with simple structures (FCC and/or BCC) [3–7]. In contradistinction to traditional alloys, four main phenomenon are conclusive to express the physical metallurgy of HEAs: the high-entropy effect, the sluggish diffusion effect, the severe lattice distortion effect and the cocktail effect [5]. HEAs are designed to form single structure in a solid solution phase, which theoretically has the minimum Gibbs free energy. This can be provided by maximizing the entropy of mixing (ΔS_{mix}). According to the high-entropy effect phenomenon, the configurational entropy (ΔS_{conf}) is the basis of ΔS_{mix} and must be higher than 1.5 R [8]. A parameter symbolized as Ω is offered to define the enthalpy – entropy relationship during solidification. The Ω -parameter is calculated as $\Omega = T_m \Delta S_{\text{conf}} / |\Delta H_{\text{mix}}|$, where T_m and ΔH_{mix} represent the melting temperature and the enthalpy of mixing, respectively. In the case of $\Omega > 1$, the formation of a solid solution phase is expected in the alloy structure. While the formations of intermetallic compounds or phase separations are expected, as in the case of $\Omega \leq 1$. According to this equation, the phase separations and the formation of intermetallic compounds are possible with decreasing temperature [9–11].

The AlCoCrFeNi system is one of the most investigated HEAs. There are many studies in the literature investigating the mechanical, microstructure, thermal, tribological and corrosion behaviors of HEAs produced by various production methods in different compositions [12–21]. Besides these studies, the effects of additional elements such as Si, Ti, C, Zr, V, Cu or Mo on the different properties of HEAs were also examined [22–33]. For instance, Yu et al. studied on the AlCoCrFeNi system and applied a heat treatment procedure. They found that some of the as-cast phases in HEA were dissolved and new phases were precipitated at different temperature ranges. Beside, it was observed that the addition of Ti increased the stability of the dendritic structure. Homogenization and recrystallization processes of Al_{0.45}CoCrFeNi HEAs were conducted by Hou et al. They examined the changes in grain size and the mechanical properties of HEAs depending on the heat treatment temperatures. The optimum mechanical properties were obtained by a heat treatment process at 800 °C for 2 h [34]. In another study on a similar alloy composition, Hou et al. performed a cold rolling process after vacuum arc melting. Then, annealing and aging processes were applied on HEAs and mechanical and microstructural properties of HEAs were investigated. They found the optimum annealing process parameter to be 900 °C for 1 h [35]. In another example, Guo et al. studied the mechanical properties and corrosion resistance of heat treated AlCoCrFeNiTi_{0.5} HEAs. The highest compression strength, strain rate and Vickers hardness were obtained after heat treatment at 1100 °C for a 6 h duration. However, the corrosion resistance of HEAs was reduced when the temperature of the heat treatment increased [36]. There have been several papers about the AlCoCrFeNiTi_x based system published during last five years. HEAs having Ti_{0.5} and Ti_{1.0} stoichiometries were the most studied subjects. The mechanical properties, corrosion behaviors, thermal stabilities and microstructural properties of these HEAs were examined under varied production procedures [22,25,31,32,36–44].

Thermo-Calc was developed by Thermo-Calc Software AB in Sweden. Phase equilibrium, phase diagram, phase transformation, and thermodynamic characteristics are all calculated using this software. Thermo-Calc was designed for complex multi-component systems with non-ideal solution phases. Thermo-Calc generates databases using the Calculation of Phase Diagram (CALPHAD) method. There are over 40 available databases studying thermodynamic, kinetic, and properties. Thermodynamic properties of pure substances, compounds, solution phases, and chemical reactions are calculated using Thermo-Calc software and its databases, as well as thermodynamic properties of aqueous solutions across a wide range [45,46].

In the literature, there are phase diagram drawings for some HEA systems using different thermodynamic software and databases [21,47]. There is some lacking in the phase diagram of the AlCoCrFeNiTi_x system which poses a problem for predicting the production conditions and the final properties of these HEAs. In order to solve this problem, in present study, several thermodynamic and kinetic calculations were carried out under selected conditions using Thermo-Calc software. Research to date has not yet been determined for phase diagram of AlCoCrFeNiTi_x HEAs. This study combines both theoretical calculations and experimental investigations and seeks to obtain data that will help to define the phase behaviors in AlCoCrFeNiTi_x HEA system.

In this study, AlCoCrFeNiTi_x based HEAs with varying stoichiometries were produced by the vacuum arc melting (VAM) process followed by heat treatments. Thermodynamic calculations were made under equilibrium and non-equilibrium cooling conditions. The phase diagram proposals have been made for these two conditions, and the accuracy of these phase diagrams has been investigated with the experimental studies and literature findings.

2. Materials and methods

The high-entropy alloys based on AlCoCrFeNiTi_x (X = 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0) stoichiometry were produced by using the Edmund Buehler AM-200 vacuum arc melting (VAM) furnace. High purity metal powders (≥ 99.5 mass %) having a fine-grain size (lower than 75 μm) were used as raw materials. Each sample was weighed as 5 g and the powder mixtures homogeneously mixed. Then, the powder mixtures were pelletized into 20 mm diameter pellets by a hydraulic press. The melting process was carried out under argon atmosphere in a water-cooled copper mold. In order to ensure chemical homogeneity, all samples were remelted by inversion five times. Five samples were melted for each alloy, to obtain repeated results. After melting process, metal ingots were cut into slices with a thickness of approximately 1.0 mm. The heat treatment (H-T) procedures were realized in a resistance furnace with the heating rate of 10 °C/min. Among all samples, heat treatments conducted in sealed evacuated quartz capsules were applied to AlCoCrFeNiTi_x based HEAs with X = 0.0, X = 0.2, and X = 1.0. The samples were kept at 500 °C, 750 °C, and 1000 °C for two hours in the Protherm PLF 150/15 chamber furnace with a heating rate of 10 °C/min. While the samples kept at 500 °C and 750 °C were cooled only in water, the samples kept at 1000 °C were cooled both in water and the furnace atmosphere to room temperature.

Different characterization procedures were applied to these bulk samples. The X-ray diffraction pattern (XRD) was achieved by using the Rigaku ULTIMA 3–Rint 2200/PC system with Cu-K α radiation (λ -K α =1.54 Å) over the range of 20–80°. The XRD results were examined by X'pert High-Score Plus software with ICDD and ICSD database. The microstructures of the HEA samples were investigated using a JEOL-JSM 6060 scanning electron microscope (SEM). The thermal analysis were made by using Perkin Elmer STA 8000 TG/DTA and LABSYS evo DTA/DSC. Samples were heated from room temperature up to 1200 °C with a heating rate of 10 °C/min under N₂ atmosphere to avoid oxidation effects. Thermo-Calc® software and TCHEA4 (High Entropy Alloys v4.0) database were also used to make thermodynamic predictions about phase stabilities, phase compositions and phase diagrams under both equilibrium and non-equilibrium cooling conditions.

3. Results and discussion

3.1. Thermodynamic predictions

The first series of thermodynamic predictions were carried out under equilibrium cooling conditions. The possible phases and their

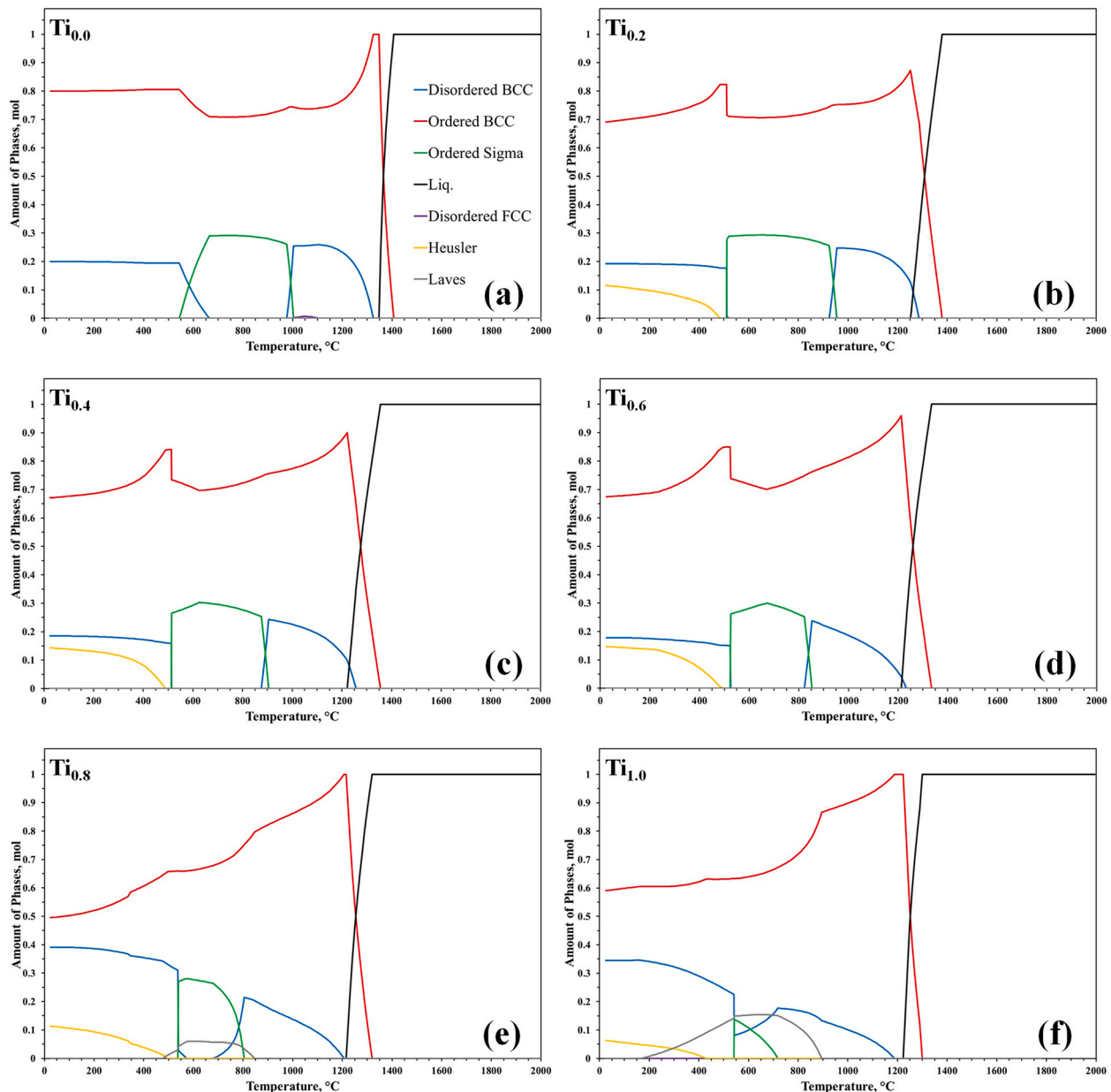


Fig. 1. The possible phases and their amounts in the microstructure of AlCoCrFeNiTi_x HEAs where a) X = 0.0, b) X = 0.2, c) X = 0.4, d) X = 0.6, e) X = 0.8, and f) X = 1.0 obtained under equilibrium cooling conditions.

amounts formed under equilibrium cooling conditions, starting from 1 mol of liquid AlCoCrFeNiTi_x based HEAs, are demonstrated in Fig. 1. During the thermodynamic investigations made with Thermo-calc software, the ratios and compositions of the theoretical phases formed with changing temperature were also calculated. Average composition values were calculated with compositions of the elements within the range of the stable temperatures of phases.

As seen in Fig. 1, formation of various phases is possible in the structure of HEAs. Under equilibrium cooling conditions, the major phase in the microstructure of HEAs can be expected to form is the ordered BCC phase (B2). The other possible phases in the structures of HEAs are the disordered BCC phase, the disordered FCC phase (only seen in Fig. 1.a), the ordered Sigma phase (intermetallic σ -phase), the L₂₁ structure of X₂YZ type full-Heusler phase, and the C14 type hexagonal Laves phase (Fig. 1e, and f). According to the thermodynamic predictions, only phases having BCC structures (ordered and disordered) can be obtained in the microstructure of AlCoCrFeNiTi_x HEAs where X = 0.0. This kind of microstructure was

also observed in the study of Ferrari et al. [33]. As can be seen in Fig. 1, it is possible that some additional phases (Heusler and Laves) will occur at certain temperature ranges with the increase of Ti content. In addition, the compositions and the stability temperature ranges of the existing phases can also change by the increase in Ti content. For example, in Ti_{1.0} alloy, there are two B2 phases (B2#2 and B2#4) which solidifies from the liquid state, and their Ti content varies between 12.91–25.83 mol% and 2.8×10^{-4} – 3.4×10^{-3} mol%, respectively. The average Ti content in the B2 phases was theoretically calculated to be 18.1 mol%. In the Ti_{1.0} alloy, two ordered BCC (B2) and two disordered BCC phases, having different compositions are likely to form during equilibrium solidification. During the equilibrium solidification, ordered B2#2 phase (Al-Co-Ni-Ti based) is formed firstly at 1306 °C. And as the temperature decreases, disordered B2 phase (Cr-based), disordered B2#3 phase (Fe-based) and ordered B2#4 phase are occurred at 1188 °C, 543 °C, and 102 °C, respectively. In Fig. 1, the sums of phases having the same crystal structure but different compositions are given. By performing XRD

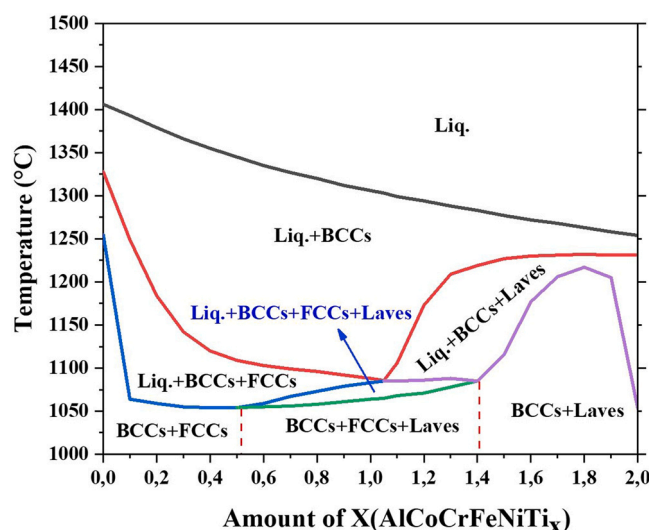


Fig. 2. The estimated phase diagram of AlCoCrFeNiTi_x HEAs plotted according to the Scheil Solidification Simulations (without any fast diffusers) under non-equilibrium conditions.

and SEM/EDS analyzes, the phases in the microstructure of HEAs can be determined more clearly.

It is known that the solidification rates cannot be controlled due to the nature of VAM process, so, alloys are obtained under non-equilibrium conditions. Therefore, Scheil Solidification Simulations should be performed to define non-equilibrium transformations and possible phase formations. Using the Scheil Calculator, the determination of the microstructure of a material, its solidification range, the level of segregation, etc., can be acquired more correctly. The estimated phase diagram calculated by the Scheil Calculator under non-equilibrium cooling conditions is shown in Fig. 2.

BCCs indicated in Fig. 2 represent the mixture of ordered BCC (B2) and disordered BCC phases with different compositions, while FCCs denote the disordered FCC and ordered FCC (L1₂) phases with different compositions. As seen in Fig. 2, the temperature where the solidification begins decreases as the Ti content in HEAs increases. It seems that, Laves phases may form during solidification when the Ti content is above approximately 0.5 ($X \geq 0.5$). In addition, there are two different forms of Laves phases: C15 (cubic structure) and C14 (hexagonal structure). The C15 Laves phase can form in HEAs where the Ti content is above 0.5, while the C14 Laves phase can form where the Ti content is above 1.2. An eutectic transformation can be observed under non-equilibrium cooling conditions. The eutectic transformation is defined as Liquid = FCCs + Laves reaction and realized approximately at 1085 °C where $X \approx 1.05$.

3.2. As-cast HEAs

The XRD patterns of AlCoCrFeNiTi_x alloys are shown in Fig. 3. The peaks having the highest intensities of ordered BCC (B2), disordered BCC (BCC) and disordered FCC (FCC) phases are diffracted at very close angles (43 – 45°) to each other. As the containing elements and their mass % change in solid solutions, shifts occur in the peaks. In order to determine the presence of these three phases, instead of looking at the highest intensity peak, the peaks diffracted from the specific planes at certain angle values were examined. The presence of the B2 phase was detected with the {100} peak at approximately 31.0°, while the {200} peak at approximately 51.0° was used for the FCC phase. If there was no reflection at these two angle values, it was determined that there is a BCC phase in the microstructure. The space groups of the selected patterns were also used to determine the possible phases. The patterns in the Pm $\bar{3}$ m, Im $\bar{3}$ m, and Fm $\bar{3}$ m

space groups were accepted to be the B2, BCC, and FCC phases, respectively. In a Ti-free alloy, a single-phase solid-solution of the B2 structure was achieved. It was observed that amount of phases were increased with increases in the Ti contents in the HEAs. There was no displacement corrections made in the XRD results. However, with the increase of Ti content in the alloy, shifts towards lower angles were observed in the peaks of the obtained phases (especially B2 phase). The reason for these shifts is that Ti has a larger atomic radius (2.15×10^{-10} m) than other constituents (1.24×10^{-10} – 1.43×10^{-10} m), which plays a part in the solid solution phases and causes the distortions of the unit cells [47]. As seen in Fig. 3b, the 2 θ position of the highest peak {110} of the B2 phase slightly shifted towards lower angles (starting from 44.68° to 43.64°) with the increase in Ti contents. These peak shifts have also been found in many studies [24,25,30]. The Sigma phase was only found in the AlCoCrFeNiTi_{0.6} alloy. It was also seen that an intermetallic Laves phase with AB₂ stoichiometry was formed in AlCoCrFeNiTi_x alloys where X equals to 0.8 and 1.0. The Laves phase contains mostly Fe, Ti and Co elements in its composition. As the amount of Laves phase formed in the structure increases, the Ti content in the B2, BCC and FCC phases decreases. The peak shift towards the high angle observed in the Ti1.0 alloy is thought to be due to this reason.

According to Fig. 2, the Laves phases may occur under non-equilibrium conditions where the stoichiometric amount of Ti is above 0.5. However, the Laves phase was not detected in the XRD result of HEA where X = 0.6 (Fig. 3b). It is thought that this situation occurs because the amount of the Laves phase formed in the structure is relatively less than the other phases. Similar to the Laves phase, FCC phases can also appear to form at lower Ti contents during non-equilibrium cooling conditions (Fig. 2). Although FCC phases are not expected to form during equilibrium cooling (Fig. 1), the appearance of these phases in the XRD results is due to non-equilibrium solidification of HEAs. It is thought that there are two reasons for the detection of the Sigma phase in the XRD analysis of HEA where X = 0.6. The first of these may be due to the phase transformation observed during equilibrium solidification (Fig. 1d). The other reason was found by using Scheil Calculator where Ni was selected as the fast diffuser. In those conditions, the formation of the Sigma phase was observed at 1099 °C. However, considering the literature studies on the diffusion of elements in HEAs, it was seen that Ni had the lowest diffusion coefficient in this HEA system [38,48,49]. For this reason, Ni cannot be a fast diffuser element. The formation of the Sigma phase can be considered by using two basic parameters: the valance electron concentration (VEC) and paired sigma-phase forming elements (PSFE). Based on these two parameters, the behaviors of HEAs can be partitioned into three groups [50,51]. The VEC and PSFE values of Ti0.6 alloy were calculated as 6.86% and 35.71%, respectively. According to these two values, Ti0.6 alloy is included in the intermediate stage alloy group. The Sigma phase may or may not occur in these alloys [51]. According to this information, the formation of the Sigma phase can be explained as the decomposition of some of the primary formed B2 and BCC mixture phases into disordered FCC and the Sigma mixture phases, during solidification.

The back-scattered electron (SEM-BE) images of AlCoCrFeNiTi_x alloys are given in Fig. 4. A homogeneous like microstructure was observed in Ti0.0 containing HEA. The morphology of this Ti0.0 alloy was found to have a single-phase structure, which was also compatible with its XRD result (Fig. 3). Similar micrographs were seen in the study by Zhou et al. [23]. The amount of phases and phase morphologies were changed with the increases in the Ti contents in HEAs. The eutectic-like structures were detected in AlCoCrFeNiTi_x alloys where X equals 0.4 and above.

According to the phase diagram proposed in Fig. 2, eutectic transformations can be observed in AlCoCrFeNiTi_x based HEAs with a Ti content between X = 0.5 and X = 1.4. The reason why an eutectic-

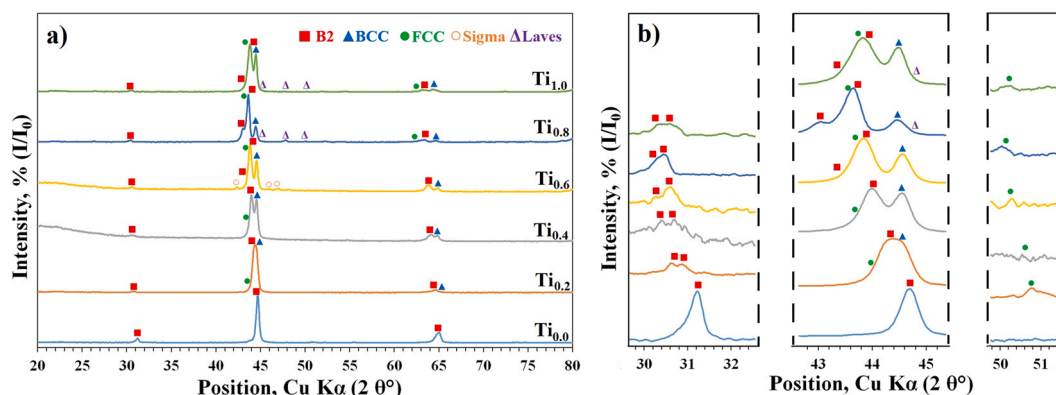


Fig. 3. XRD analysis results of as-cast AlCoCrFeNiTi_x based HEAs (a) full scan, (b) magnified sections.

like microstructure was observed (Fig. 4.c) in the microstructure of HEA where $X = 0.4$, which is outside these limits, was attempted to be determined by using Scheil Solidification Simulation. Using the Scheil Calculator, the formation of Laves and FCC phases from the liquid phase was observed at 1058 °C, where Ti was selected as the fast diffuser. Nevertheless, considering again the literature studies

on diffusions of elements in HEAs, it has been stated by Chen et al. that Ti has a lower diffusion coefficient than Fe, Al, Cr, and Co in this HEA system [38]. So, Ti cannot be a fast diffuser element in this system. There are several reasons for the formation of different phases in the as-cast alloys structure according to the Scheil calculation. These reasons are as follows: diffusion in the liquid is not fast

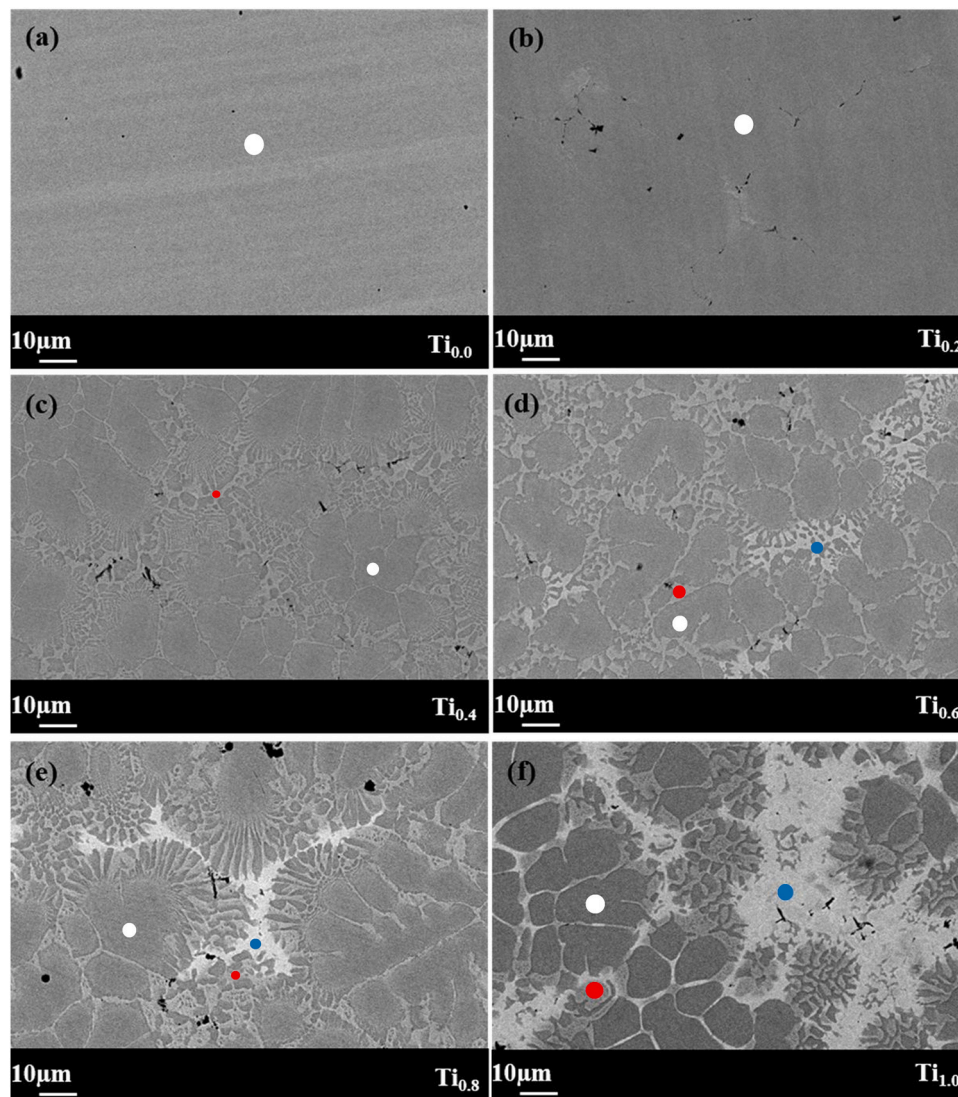


Fig. 4. SEM-BE micrographs of as-cast AlCoCrFeNiTi_x based HEAs where (a) $X = 0.0$, (b) $X = 0.2$, (c) $X = 0.4$, (d) $X = 0.6$, (e) $X = 0.8$, and (f) $X = 1.0$.

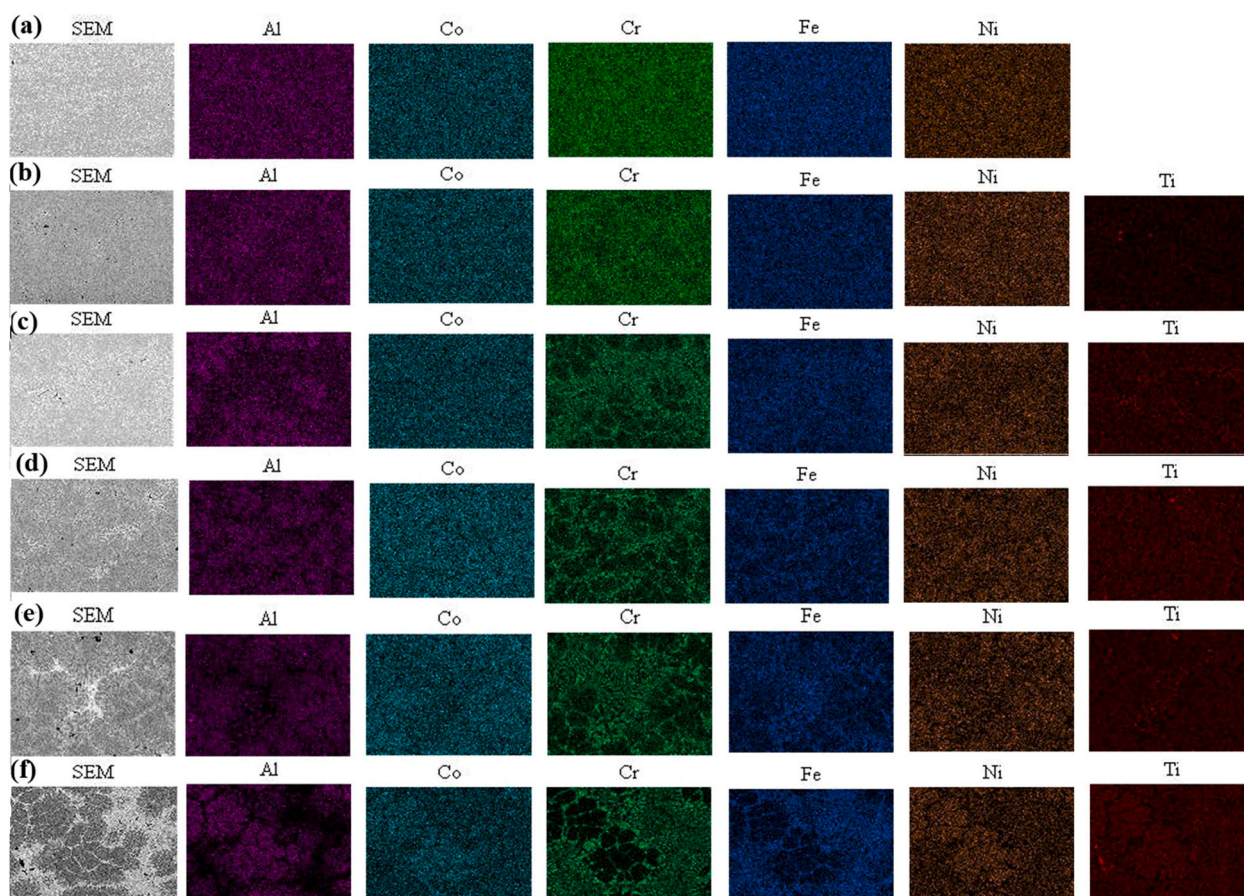


Fig. 5. X-ray mapping of the elements in as-cast AlCoCrFeNiTi_X based HEAs where (a) X = 0.0, (b) X = 0.2, (c) X = 0.4, (d) X = 0.6, (e) X = 0.8, and (f) X = 1.0 (1000 magnification).

enough, back diffusion occurs in the solid phases, and absence of the possible phases due to high nucleation energy [47]. In order to better understand the element distributions in AlCoCrFeNiTi_X based HEAs, EDS X-ray mapping studies were carried out and given in Fig. 5.

As seen in Fig. 4 and Fig. 5, the dendritic structures were detected in the microstructure of AlCoCrFeNiTi_X based HEAs where X equals 0.4 and above. The formations of the dendritic structures can occur due to the solidification after arc melting. In addition to this, some casting cavities are also seen as black regions (Fig. 4) in the microstructures of HEAs. As seen in Fig. 5, the compositions of Al, Ni and Ti in the dendrite structure were measured at higher concentrations than the interdendritic region. In the interdendritic region, eutectic-like structures were seen. Also, higher concentrations of Cr, and Fe were measured in these regions. On the other hand, it was observed that the Co element was spread uniformly across the microstructures. Due to the rapid solidification, these kind of non-homogeneous dendritic structures were obtained. This behavior was also seen in the study of Ham et al. [25]. Similar microstructure images were also seen in several studies [22,25].

The compositions of the phases calculated by Thermo-Calc software under both equilibrium and non-equilibrium cooling conditions were compared with EDS analysis results. For example, the different regions were selected for AlCoCrFeNiTi HEA and are indicated in Fig. 4f. The EDS result of the first solidified phase (white point) coincides with the composition of the ordered + disordered mixture structure (BCC_B2) determined with the Scheil calculator. The phases in the microstructures given in Fig. 4 marked with the

white points belong to B2 and BCC phase mixtures. As regards the Scheil calculator, a peritectic transformation should occur at a temperature of 1211 °C. The peritectic reaction follows as Liquid + BCC_B2 = BCC_B2#2. These two BCC phases have the mixture of both ordered and disordered structures comprising of different compositions. During this transformation, it is expected that Cr and Fe will increase while Al and Ni decrease in the composition of the newly formed phase. These composition differences were detected by EDS analysis of two separate regions indicated in Fig. 4f. The red pointed phase was thought to belong to the BCC_B2#2 phase. According to the Scheil calculator, an eutectic transformation is expected to occur at 1087 °C for AlCoCrFeNiTi HEA. The theoretical composition of the liquid at this temperature is predicted to be 1.22 at% Al, 29.07 at% Co, 19.17 at% Cr, 27.80 at% Fe, 6.84 at% Ni and 15.88 at% Ti, respectively. The result of the EDS analysis of the eutectic region (blue point) were measured as 1.88 at% Al, 13.60 at% Co, 27.29 at% Cr, 30.61 at% Fe, 9.88 at% Ni and 14.97 at% Ti, respectively. Similar compositions were obtained with the Scheil calculator. The main differences observed in the composition of Co. The phases (Laves and disordered FCC) occurred due to the eutectic reaction were also detected in the XRD analysis (Fig. 3). The blue marked points (Fig. 4) belong to the Laves and disordered FCC phase mixtures (eutectic phases).

DTA analyzes were carried out to determine the temperatures at which heat treatment (H-T) procedures will be carried out. The DTA analysis results of the as-cast HEAs are shown in the Fig. 6. The phase transformations are critical for comprehending the different

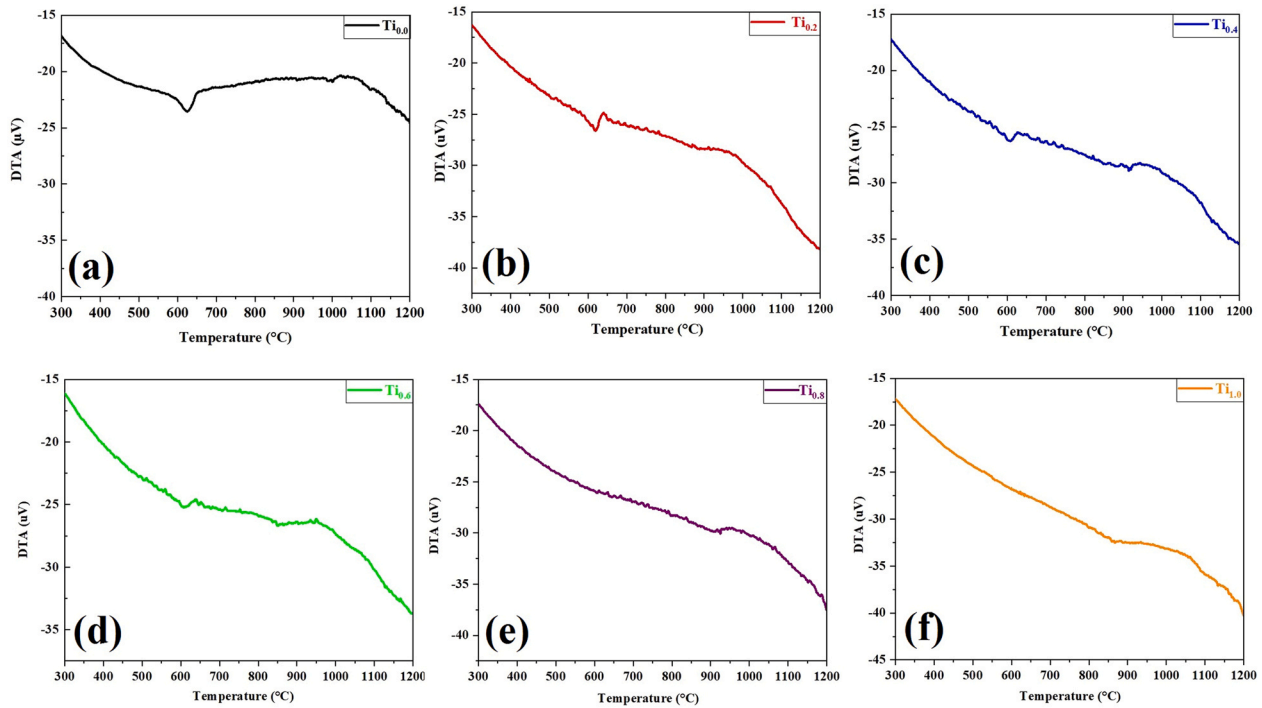


Fig. 6. DTA analysis of as-cast AlCoCrFeNiTi_x based HEAs where (a) X = 0.0, (b) X = 0.2, (c) X = 0.4, (d) X = 0.6, (e) X = 0.8, and (f) X = 1.0.

phases that may exist on a given microstructure. The endothermic peaks were observed in all HEAs approximately at 600 °C and 900 °C. The operating temperatures of H-T process were chosen below and above these two transformations.

3.3. Heat treated HEAs

In relation to the DTA analysis results, three different heat treatment temperatures were selected: 500 °C, 750 °C, and 1000 °C. The XRD patterns of the heat treated AlCoCrFeNiTi_x alloys are shown in Fig. 7.

As seen in Fig. 7a, and b, there was no phase changing detected between AC alloys and HT alloys at 500 °C. However, the peaks of the Laves phases in the AC alloy of AlCoCrFeNiTi were not detected in the XRD results of HT alloys at 500 °C. It was observed that, with the increase in Ti content, BCC and FCC phases were formed in addition to B2 phases in HT alloys at 500 °C. The Sigma phase was formed during the H-T procedures at 750 °C for all HEAs. This indicates that the first endothermic peaks detected in the DTA analysis (Fig. 6) belong to the formation of the Sigma phases. The XRD results of

HEAs heated to 1000 °C and cooled in two different environments were obtained similar to each other. These changes observed in XRD results should be evaluated with SEM and EDS analysis. The microstructures of HEAs obtained after H-T at different temperatures and cooling conditions are given in Fig. 8.

Similar to the XRD results (Fig. 7), SEM and EDS analysis of HT alloys, heated at 500 °C followed by water cooling (Fig. 8.iv), and AC alloys were comparable. The Sigma phases were also detected in the SEM microstructures of HEAs heat treated at 750 °C followed by water cooling (Fig. 8.iii). EDS analysis of the light colored regions demonstrated that the composition of these regions were close to theoretical compositions of Sigma phases obtained under equilibrium conditions. For example, the chemical composition of the light colored region in HT AlCoCrFeNi alloys at 750 °C was measured as 9.41 at% Co, 44.76 at% Cr, and 37.85 at% Fe, respectively. While, the average of the theoretical composition of the Sigma phase obtained by equilibrium cooling conditions is 5.75 at% Co, 53.46 at% Cr, and 40.24 at% Fe, respectively. The microstructures of HT HEAs at 1000 °C both water and furnace cooled looked similar. There were some precipitations and eutectoid-like phases identified in the

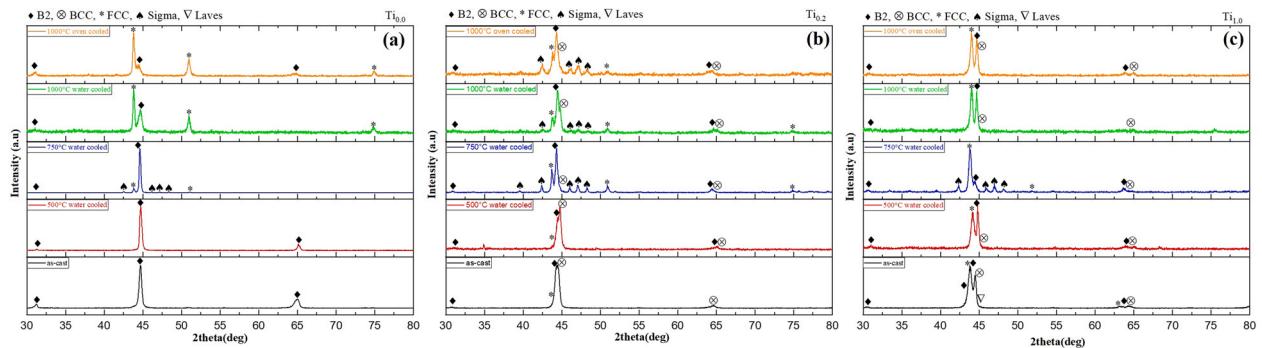


Fig. 7. XRD analysis of heat treated AlCoCrFeNiTi_x based HEAs where a) X = 0.0, b) X = 0.2, and c) X = 1.0.

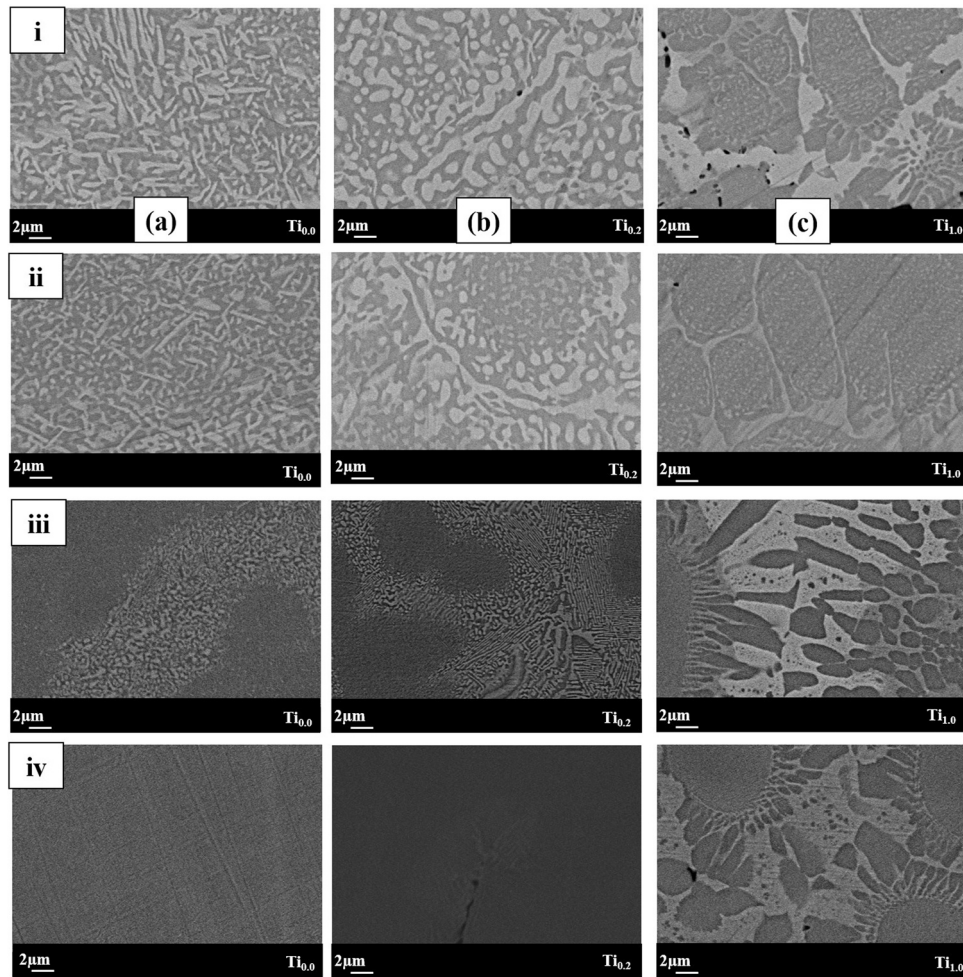


Fig. 8. SEM-BE micrographs of heat treated AlCoCrFeNiTi_X based HEAs where (a) X = 0.0, (b) X = 0.2, and (c) X = 1.0. H-T conditions: (i) at 1000 °C – cooled in furnace, (ii) at 1000 °C – cooled in water, (iii) at 750 °C – cooled in water, and (iv) at 500 °C – cooled in water.

microstructure of HT AlCoCrFeNiTi alloys (Fig. 8c.i, and c.ii). A comparison of the DSC analysis and the obtained microstructures after H-T is given in Fig. 9.

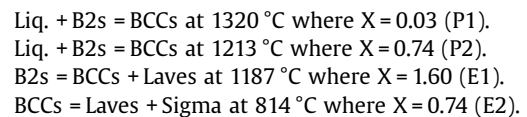
The transformations that occurred during the heating and cooling of the as-cast HEAs are given in Fig. 9. The effect of the H-T process realized at the temperatures below and above the transformations on the microstructure is also given in Fig. 9. Titanium addition affects the casting microstructure and different phases form. Depending on the composition of these phases in the casting, the phase transformation temperatures also change. Differences in phase transformation temperatures were also seen in the DTA (Fig. 6) and DSC (Fig. 9) analysis. While the first fundamental phase transformation temperature was determined as 626 °C in the Ti_{0.0} alloy, this transformation temperature was observed to be 556 °C in the Ti_{1.0} alloy. This shows that the heat treatment temperature changes with the casting microstructure. In all three HEAs, peaks corresponding to exothermic transformations were observed during cooling at similar temperatures against the endothermic transformations observed during heating.

3.4. Phase diagram proposal

The phase equilibrium diagram calculated by Thermo-Calc software for the AlCoCrFeNiTi_X based system, which includes the

transformations under equilibrium cooling conditions is given in Fig. 10. Phases donated by B2s and BCCs represent the sum of the ordered BCC and disordered BCC phases having different compositions, respectively. Although the transformations of the B2s and BCCs phases were not clearly seen in DTA and DSC analyzes, the formation and dissolution of the Sigma phase in HEAs were determined more clearly.

As seen in Fig. 10.a, as the Ti content in HEAs increases, the theoretical melting temperature decreases. The firstly solidified phase in HEAs is the ordered structure of BCC (B2) phase. When the Ti content is above 1.6, the second solidifying phase is the Laves phase, while when below this composition amount, the disorder structure of BCC phase solidifies secondarily. It is predicted that there may be two peritectic and two eutectoid transformations in the phase diagram. The reactions are as follows:



The findings of the previous studies and the thermal analyzes performed in this study are shown in Fig. 10b. The temperature differences between liquidus and solidus were found to be higher in

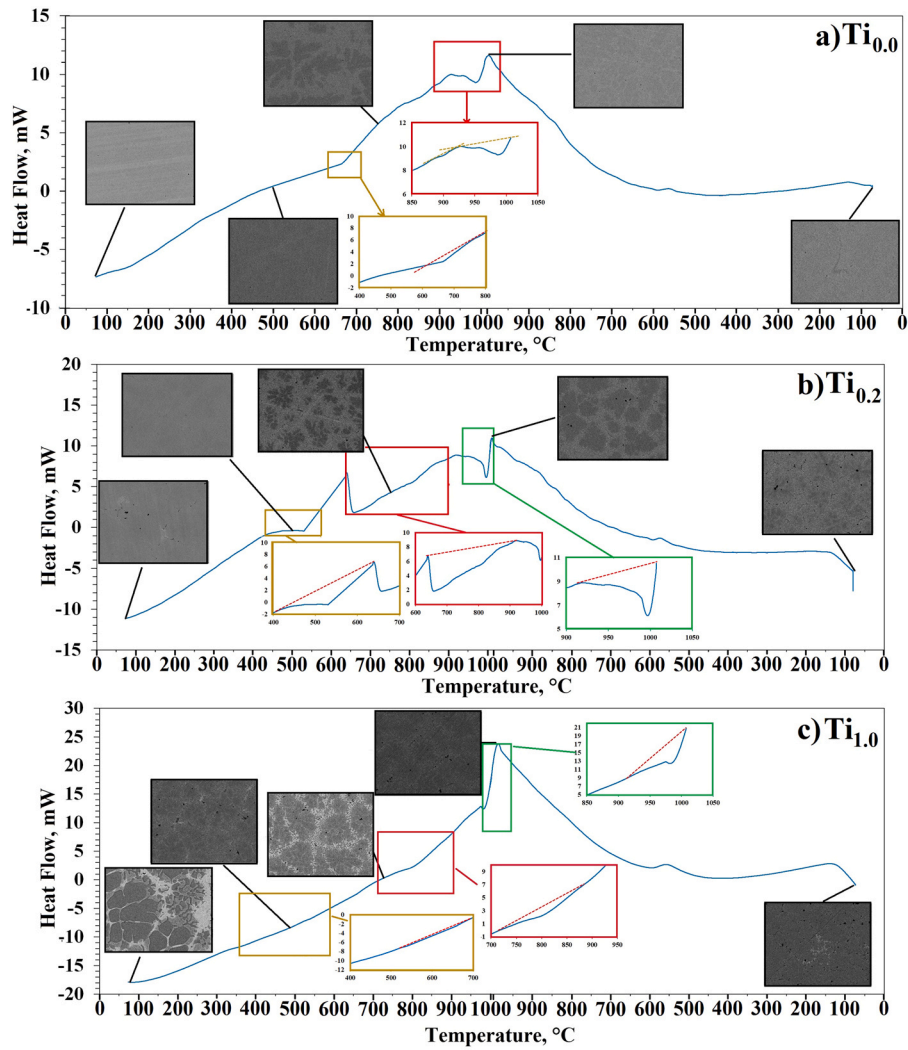


Fig. 9. The results of DSC analysis and the microstructures of AlCoCrFeNiTi_x based HEAs where (a) X = 0.0, (b) X = 0.2, and (c) X = 1.0 obtained by H-T processes.

experimental studies in the literature [22]. As for the DTA and DSC results, there were lots of temperature peaks determined for possible phase transformations. The most prominent ones of these transformations were selected and the phase transformation

temperatures are presented in the Fig. 10b. Similarities were observed between temperatures both experimentally measured and theoretically calculated that show the stability region of the Sigma phase.

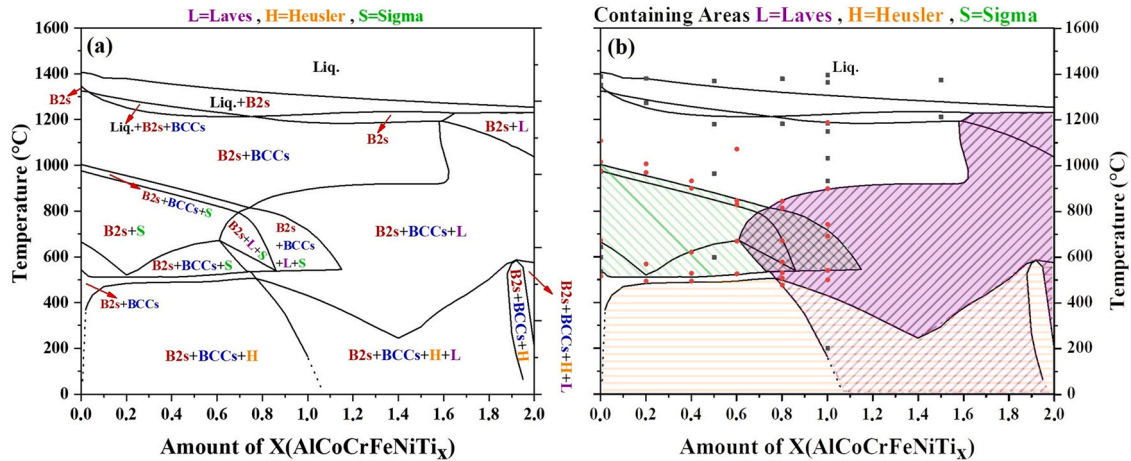


Fig. 10. Phase equilibrium diagram of AlCoCrFeNiTi_x based HEAs plotted by (a) Thermo-Calc software calculations, and (b) ■ = literature studies [22,36,39,52–55] with • = experimental findings.

4. Conclusion

In summary, vacuum arc melting process was utilized to produced AlCoCrFeNiTi_x (X = 0.0, 0.2, 0.4, 0.6, 0.8, and 1.0) based HEAs. The heat treatment processes at various temperatures (500 °C, 750 °C, and 1000 °C) and cooling conditions (water and furnace atmosphere) were followed after the VAM process. The CALPHAD method was also used to plot the phase diagrams of this HEA system under both non-equilibrium and equilibrium cooling conditions and the results were compared with the experimental findings.

- It was seen in the XRD results of as-cast HEAs that a single phase ordered BCC solid solution was detected in the AlCoCrFeNi alloy, and with the increase in Ti content, disordered BCC and disordered FCC phases were also formed. In addition to these phases, Laves and Sigma phases with low intensities were also detected.
- Similar results with the XRD analyzes were also obtained in SEM microstructures. A single matrix phase was observed in the AlCoCrFeNi alloy. As the Ti content increased, formations of multiple phases with different contrasts were formed. The eutectic-like structures (X = 0.4, 0.6, 0.8, and 1.0) and the peritectic-like structures (X = 0.8 and 1.0) were observed in the microstructures. According to the EDS results, the composition of the firstly solidified phase matched the composition of the ordered BCC and disordered BCC mixture phases. The composition of the peritectic-like structure belonged to the ordered BCC and disordered BCC mixture phase which have different compositions. Also, the composition of the eutectic-like structure complied with the composition of Laves, disordered FCC and disordered BCC mixture phases.
- These eutectic and peritectic transformations coincide with the phase diagram calculated with the Scheil Calculator which was generated under non-equilibrium cooling conditions.
- There were endothermic peaks observed in the DTA results of all HEAs. According to the DTA results, the heat treatment experiments were conducted at 500 °C, 750 °C, and 1000 °C. Considering the XRD analyzes of HT alloys, there were no remarkable phase changes in the structure of HEAs as a results of H-T at 500 °C. For all HEAs, the Sigma phase was generated during the H-T procedures at 750 °C. This shows that the first endothermic peak found in the DTA analysis is associated with the formation of the Sigma phases.
- Similar to the XRD results, SEM and EDS analysis of HT alloys, heated at 500 °C followed by water cooling, were comparable to AC alloys. The Sigma phases were also detected in the selected region in the SEM microstructures of HEAs H-T at 750 °C. EDS analysis demonstrated that the composition of these regions were closed to theoretical compositions of the Sigma phases obtained under equilibrium conditions. The microstructures of HT HEAs at 1000 °C both water and furnace cooled looked alike. There were some precipitations and eutectoid-like phases that were identified in the microstructure of HT AlCoCrFeNiTi alloys.
- The phase equilibrium diagram calculated by the CALPHAD method under equilibrium cooling conditions was plotted. The predictions for the phase transformations were made with the help of this phase diagram. The findings of both the previous studies and present study were compared with the phase diagram proposal. It was determined that the formation and dissolution temperatures of the Sigma phases are in good agreement with the theoretical calculations.

CRediT authorship contribution statement

Saadet Güler: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original

draft, Writing – review & editing, Visualization. **Esra Dokumaci Alkan:** Conceptualization, Methodology, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition. **Murat Alkan:** Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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